

UTSAV CHAUDHARY

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Senior **Full Stack Engineer** with **7+ years** of hands-on **backend** and **frontend** development, plus **AI agents, bots, and LLM** systems. Expert in **Python, TypeScript, and JavaScript** with **Angular, React.js, and Next.js**, and in **REST APIs** using **FastAPI, Flask, and Node.js**. Provide **solution design**, development quality, and delivery execution on **AWS** and **Azure** (compute, storage, networking, platform services). Guide teams on **CI/CD, Git workflows, Terraform, and GitHub Actions**, with consistent **SOLID, automated testing, coding standards, and secure coding**.

Technical Skills

Languages & Frontend: Python, TypeScript, JavaScript, C#, Angular, React.js, Next.js, Redux, TailwindCSS, D3.js, Chart.js

Backend & APIs: FastAPI, Flask, Django, ASP.NET, ASP.NET Web API, Node.js (Express), Pydantic, REST APIs, OpenAPI, Swagger, gRPC, WebSockets, GraphQL

AI, LLMs, Agents, & Bots: Azure OpenAI, GPT-4o, GPT-4o-mini, Llama, Hugging Face, RAG, embeddings, vector search, MCP servers, AI agents, Ollama, Whisper, TensorFlow Lite, prompt engineering

Cloud, DevOps, & CI/CD: AWS (EC2, EKS, Lambda, RDS, S3, VPC, IAM, API Gateway, CloudWatch), Azure (App Services, AKS, Azure DevOps, Azure SQL, Application Gateway, Key Vault), Docker, Kubernetes, Terraform, Ansible, Jenkins, GitHub Actions, Bitbucket Pipelines, Git, GitFlow, environment promotion

Networking & Security: VNets, Security Groups, NSGs, DNS, VPN, LDAP, RBAC, SSL, TLS, OAuth 2.0, HIPAA, GxP, secure coding, vulnerability assessments, enterprise risk controls

Engineering Practices & Tools: OOAD, SOLID, automated testing, code reviews, design reviews, branching strategies, PostgreSQL, MongoDB, Redis, Kafka, Jira

Experience

Sr. Full Stack Engineer / Technical Lead, Hybrid, **Labs196, TX** Feb 2025 to Present

- Provided technical leadership on parallel AI and product workstreams, working with product and architecture leads to turn business requirements into **technical specifications** for APIs and UI delivery.
- Led **code reviews, design reviews**, and technical validations so assigned workstreams stayed consistent on **architectural principles, OOAD, SOLID, automated testing, coding standards**, versioning, and documentation.
- Designed **AI agents, bots, and MCP servers** that extended **LLM** capabilities with custom tools, file systems, and APIs; owned the manager agent that planned tasks, picked tools, and routed work to specialist agents with human in the loop gates.
- Built the shared **Python/TypeScript** tool layer (search, scrape, DB, Gmail) with schemas and retries; wired **Azure OpenAI GPT-4o/GPT-4o-mini**, and a **Hugging Face** extraction model as callable tools before cleaned records hit **MongoDB**.
- Built **React.js** and **Next.js TypeScript** applications with **TailwindCSS**, including an internal Admin Portal for KPI tracking and agent-run monitoring; translated **Figma** into production UI with **RBAC** and performance-tuned component patterns.
- Developed **Python** microservices and **Node.js REST APIs** over **MongoDB** and **Firebase**; integrated **Stripe, HubSpot**, and **Sumsb** with **PCI aligned secure coding**.
- Partnered with DevOps on **CI/CD (GitHub Actions, Bitbucket)**, **Terraform, Docker** on **EKS/AKS**, and **environment promotion** across **Dev, Test, UAT, and Prod**; led **root cause analysis** with **Prometheus** and **CloudWatch**.

Full Stack Engineer, On-Site, **Delta Dental, GA** Jan 2024 to Jan 2025

- Implemented and reviewed **Python FastAPI/Flask** and **Node.js REST APIs** for member eligibility and claims, with **automated testing** before **AKS** promotion, helping keep **99.9% availability** behind **Application Gateway** and **Front Door**.
- Delivered **Angular** and **React.js TypeScript** features for the member enrollment and provider self service portals, keeping UI routing and API contracts intact through **environment promotion** on Azure.
- Performed **code reviews** and **design reviews** on portal and API changes; applied **OOAD, SOLID**, and **OWASP** with coding standards, and **HIPAA aligned secure coding** with **Azure Key Vault** for member PHI and claims data in **Azure SQL**.
- Helped migrate the enrollment and provider portal stack (**Angular, React, Python, Node.js**) from legacy hosting to cloud native **AKS** on Azure, improving **resource utilization** by **40%** while preserving service boundaries and API contracts.
- Supported canary releases and **environment promotion** through **Azure DevOps** pipelines, **Terraform, Git workflows**, and **branching strategies**, contributing to a **75%** reduction in deployment related incidents.
- Contributed to monitoring with **Application Insights, Azure Monitor, and Grafana**, supporting **MTTD** from **30 minutes** to **5 minutes**; led **root cause analysis** and **vulnerability assessments** with Risk and Compliance.

Full Stack Engineer CO-OP, On-Site, **Midh Technologies, TX** June 2023 to Dec 2023

- Built **Python FastAPI** and **Flask REST APIs** for **Llama** and **Hugging Face** inference behind **AWS API Gateway / Azure API Management**, with **IAM** authentication, request throttling, and **Pydantic** validation.
- Developed **React.js** analytics dashboards with **TypeScript, D3.js**, and granular **RBAC** for internal teams and client stakeholders, translating analyst requirements into production UI workflows.

- Fine tuned **Llama** models on GPU backed **Kubernetes (EKS/AKS)**; contributed **Python ETL** and **Pandas** jobs that loaded curated tables into **PostgreSQL** and **MongoDB** for dashboard and API consumers.
- Built and deployed the company web platform in **Next.js** and **React** with secure **RBAC** and scalable API integrations on **Azure App Service** and **AWS Amplify**.
- Built production **CI/CD** with **GitHub Actions** and **Azure DevOps**: **pytest** unit tests, Docker image builds, security scanning, **Terraform**, and **environment promotion**.

Software Engineer, Python, On-Site, Compendious Medialabs Pvt. Ltd., IN June 2021 to July 2022

- Built a responsive **React** frontend on a **FastAPI** backend, delivering interactive **D3.js** and **Chart.js** metrics dashboards that improved data visualization efficiency by **45%**.
- Optimized **Python FastAPI REST APIs** with **Pydantic** models, sustaining **250ms** average response time behind **Nginx** with **SSL/TLS** termination.
- Built **Python** scraping/ETL with **Selenium**, **BeautifulSoup**, and **Pandas**, cutting manual collection time by **60%**; added **WebSocket** streaming so operational dashboards stayed current.
- Implemented self hosted **CI/CD** with **GitHub Actions** and **Jenkins**, **Git branching strategies**, and **Prometheus/Grafana** observability; applied **code reviews** and reduced operational overhead by **60%**.

Software Engineer, Remote, VRoKCs, MO March 2020 to May 2021

- Deployed **Python Flask** web applications and **REST APIs** on **Azure App Services** with high availability, auto scaling, and integrated monitoring across compute, storage, and networking.
- Built a **Flask** app (**Python, JavaScript**) with **TensorFlow Lite** for real time, in browser sign language model evaluation via image upload, webcam streaming, and live prediction display.
- Built **CI/CD** through **Azure DevOps** and **Git** versioning for model and application deployment, using **Azure Blob Storage** for artifacts and **Docker** for inference services, reducing deployment cycles by **40%**.
- Applied **automated testing** and **secure coding** on critical path **Flask** APIs and JS client work; partnered with ML engineers on **Python** preprocessing for data collection, annotation, and dataset preparation.

Software Developer, On-Site, Shree Hari Marketing PVT. LTD., IN March 2019 to Feb 2020

- Implemented **REST APIs** in **Python Flask** and **Node.js** for order, inventory, and customer workflows, following **REST** resource design, for maintainable service boundaries.
- Built the customer facing **Angular** and **JavaScript** UI against those APIs, and wrote **Python** scripts for reporting and data cleanup against **SQL Server**.
- Used **Git** versioning, **Jenkins** builds, and release checklists for **environment promotion** across Dev, Test, and Production.
- Performed **code reviews**, unit and integration **automated testing**, and **secure coding** checks (input validation, auth, logging) before production releases.

Web/Game Developer, On-Site, underDogs Gaming Studios, IN Jan 2018 to Feb 2019

- Built game service **REST APIs** in **Python** and **Node.js**, integrating the **Quarters API** for wallets, currency conversion, and secure transactions across multiple titles.
- Wrote **Python** tooling and **JavaScript** admin screens for payments, inventory, and shipping, keeping client/server flows reliable when requests failed or the network dropped.
- Stood up basic release and hosting for game backends, including load balancing, **API** rate limiting, health checks, and **Git** source control for client and server code.
- Applied **API** security, logging, and integration **automated testing** for payment failures and network edge cases before production game releases.

Certifications



Microsoft Certified: DevOps Engineer Expert

AZ-400: Designing and Implementing Microsoft DevOps Solutions

AZ-900: Microsoft Certified: Azure Fundamentals

Google IT Automation with Python (Specialization)

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